

Angle Measures on Parallel Lines

Answer Key

High-School Geometry Lines and Angles

Quick Answers

1. 62, 118	4. 125, 55	7. 16, 76	10. 14, 122
2. 133, 47	5. 12, 69	8. 22, 102	
3. 17, 61	6. 108, 72	9. 15, 64	

Curriculum

Level: High-School Geometry Lines and Angles

HSG-CO.C.9 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 21 tagged checks.

Common wrong answers

3. If they got 17: stopped at x and reported it as the angle measure

7. If they got 20: set the two expressions equal instead of adding them to 180, getting x of 2 and an angle of 20

8. If they got 22: stopped at x and reported it as the angle measure

Angle names refer to the shared diagram on the worksheet: $\ell \parallel m$ cut by transversal t , with a, b, c, d around the upper intersection and e, f, g, h around the lower one.

1. $m\angle a = 118^\circ$. Find $m\angle d$ and $m\angle e$.

Step 1. $\angle a$ and $\angle d$ sit on the same side of the transversal on either side of line ℓ , so together they make a straight angle — a linear pair.

Step 2. $180 - 118 = 62$, so $m\angle d$ is

62°

Step 3. $\angle e$ occupies the same position at line m that $\angle a$ occupies at line ℓ , so they are corresponding angles, and the lines are parallel — corresponding angles are therefore congruent.

118°

Listen for: “they look about the same”. The diagram is not to scale; the reason is the parallel-line postulate, not the picture.

2. $m\angle c = 47^\circ$. Find $m\angle f$ and $m\angle g$.

Step 1. $\angle c$ and $\angle f$ are both between the two parallel lines and both on the same side of the transversal — same-side interior angles, which are supplementary.

Step 2. $180 - 47 = 133$.

133°

Step 3. $\angle g$ corresponds to $\angle c$, so the two are congruent.

47°

Teaching note: same-side interior is the one pair on this diagram that adds to a straight angle rather than matching. Students who memorise “parallel means equal” get this one wrong every time.

3. $m\angle b = (3x + 10)^\circ$, $m\angle f = (5x - 24)^\circ$.

Step 1. $\angle b$ and $\angle f$ are corresponding angles, so with $\ell \parallel m$ their measures are equal. Set the expressions equal.

Step 2. $3x + 10 = 5x - 24$ gives $34 = 2x$, so

$$x = 17$$

Step 3. Substitute back into either expression: $3(17) + 10 = 51 + 10$.

$$61^\circ$$

Common wrong answer: 17 given as the angle. Finding x is only the first half; the question asked for a measure in degrees. Checking the other expression is worthwhile too: $5(17) - 24 = 61$ as well.

4. $m\angle h = 125^\circ$. Find $m\angle b$ and $m\angle e$.

Step 1. $\angle b$ and $\angle h$ are both outside the parallel lines and on opposite sides of the transversal — alternate exterior angles, which are congruent.

$$125^\circ$$

Step 2. $\angle h$ and $\angle e$ are adjacent along line m on the same side of it, so they form a linear pair.

Step 3. $180 - 125 = 55$.

$$55^\circ$$

Teaching note: alternate exterior is the pair students meet least often. Point out that it is the mirror image of alternate interior, taken outside the parallel lines instead of between them.

5. $m\angle d = (7x - 15)^\circ$, $m\angle f = (4x + 21)^\circ$.

Step 1. Both angles lie between the parallel lines and on opposite sides of the transversal — alternate interior angles, so they are congruent.

Step 2. $7x - 15 = 4x + 21$ gives $3x = 36$, so

$$x = 12$$

Step 3. $7(12) - 15 = 84 - 15$.

$$69^\circ$$

Check it: the other expression gives $4(12) + 21 = 69$ as well, which is the fastest way for a student to catch an arithmetic slip.

6. $m\angle f = 108^\circ$. Find $m\angle d$ and $m\angle a$.

Step 1. $\angle d$ and $\angle f$ are alternate interior angles, so they are congruent.

$$108^\circ$$

Step 2. There is no single named pair joining $\angle f$ to $\angle a$, so go through $\angle d$: $\angle a$ and $\angle d$ form a linear pair at the upper intersection.

Step 3. $180 - 108 = 72$.

$$72^\circ$$

Teaching note: this is the first problem that needs two relationships chained together. Writing the middle result down before reaching for the second rule is the habit worth building.

7. $m\angle c = (4x + 12)^\circ$, $m\angle f = (6x + 8)^\circ$.

Step 1. $\angle c$ and $\angle f$ are same-side interior angles, so they are supplementary, not congruent. Their sum is 180.

Step 2. $(4x + 12) + (6x + 8) = 180$ gives $10x + 20 = 180$, then $10x = 160$, so

$$x = 16$$

Step 3. $4(16) + 12 = 64 + 12$.

$$76^\circ$$

Common wrong answer: 20 — from setting the two expressions equal, which gives $x = 2$. That is the congruent rule applied to a supplementary pair. Sanity check: $76 + 104 = 180$, as a same-side interior pair must.

8. $m\angle d = (5x - 8)^\circ$, $m\angle e = (3x + 12)^\circ$.

Step 1. $\angle d$ is interior at the upper line and $\angle e$ is interior at the lower line, both on the same side of the transversal — so they are supplementary.

Step 2. $(5x - 8) + (3x + 12) = 180$ gives $8x + 4 = 180$, then $8x = 176$, so

$$x = 22$$

Step 3. $5(22) - 8 = 110 - 8$.

$$102^\circ$$

Common wrong answer: 22 reported as the angle. Check the pair: $m\angle e = 3(22) + 12 = 78$, and $102 + 78 = 180$.

9. $m\angle a = (2x + 34)^\circ$, $m\angle g = (5x - 11)^\circ$.

Step 1. $\angle a$ is outside the parallel lines at the upper intersection and $\angle g$ is outside them at the lower one, on the other side of the transversal — alternate exterior angles, so congruent.

Step 2. $2x + 34 = 5x - 11$ gives $45 = 3x$, so

$$x = 15$$

Step 3. $2(15) + 34 = 30 + 34$.

$$64^\circ$$

Teaching note: a student who cannot see the pair can still get there in two steps — $\angle a$ and $\angle c$ are vertical angles, and $\angle c$ and $\angle g$ correspond.

10. $m\angle b = (9x - 4)^\circ$, $m\angle e = (3x + 16)^\circ$ — **decide the relationship yourself.**

Step 1. $\angle b$ and $\angle e$ are not one of the named pairs, so chain two that are: $\angle b$ and $\angle f$ correspond, hence are congruent, and $\angle f$ and $\angle e$ form a linear pair. So $\angle b$ and $\angle e$ are supplementary — they add to a straight angle, they are not equal.

Step 2. $(9x - 4) + (3x + 16) = 180$ gives $12x + 12 = 180$, then $12x = 168$, so

$$x = 14$$

Step 3. $9(14) - 4 = 126 - 4$.

$$122^\circ$$

Step 4. Check: $m\angle e = 3(14) + 16 = 58$, and $122 + 58 = 180$.

Open response (manual review): the written justification is the student's own. Accept any answer that reaches supplementary through a valid chain — corresponding then linear pair, as above, or vertical angles then same-side interior. Do not accept “they looked supplementary”; the point of the question is the chain of reasons. A student who assumed congruent gets $6x = 20$ and a non-integer x , which is itself a signal that the wrong rule was chosen.